Limits @ ∞ &
The Squeeze Thm

Lecture 4: Limits III

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Quiz Time!

Let's get into formation for quiz time.

Last Time

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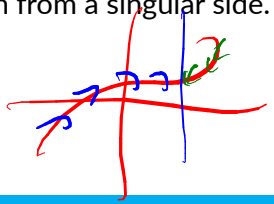
$\lim_{x \to a} f(x) = f(a)$ if $f(x)$ cont. @ a

Last Time

We spent much of last lecture developing our ability to compute limits via:

1. Recognizing when functions were *continuous* so that we could reduce the computation of limits to simpler evaluations
2. Developing limit properties which allowed us to build up the computation of complicated limits via reduction to simpler ones

We also developed the notion of the *one sided limits* - restricted kinds of limits where we need only approach from a singular side.



$\lim_{t \to \infty}$

Today's Goals

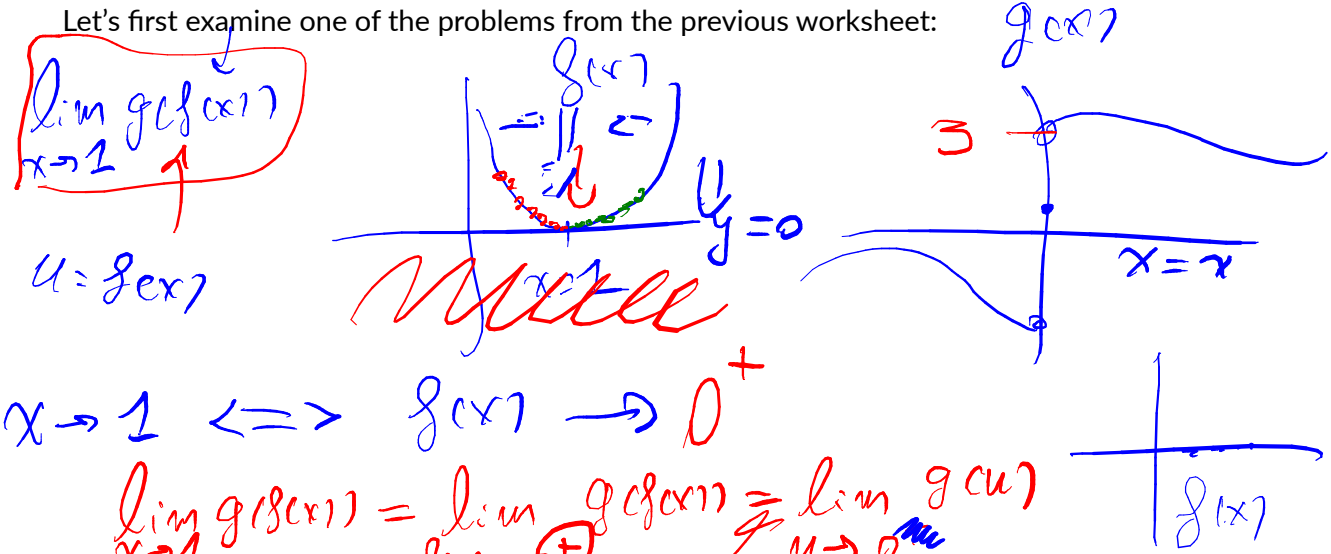
We'll apply both our new limit computation skills as well as our knowledge of one-sided limits to work with *limits at or involving infinity*.

We will also uncover our last tool to help us compute limits: *the Squeeze Theorem*.

Limits at/involving Infinity

Prelude I

Let's first examine one of the problems from the previous worksheet:



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We had two functions whose graphs looked like:

$\lim_{x \to 1} f(x) = 0$

$u = f(x)$

$x \to 1 \Rightarrow u \to 0^+$

$\lim_{x \to 1} g(f(x)) = \lim_{u \to 0^+} g(u) = 3$

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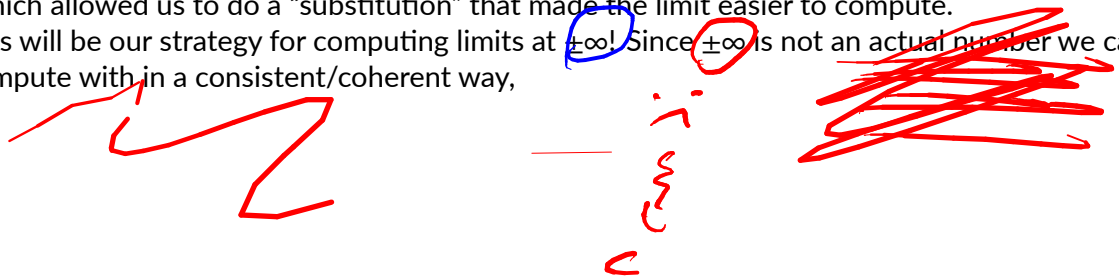
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This will be our strategy for computing limits at $\pm\infty$!

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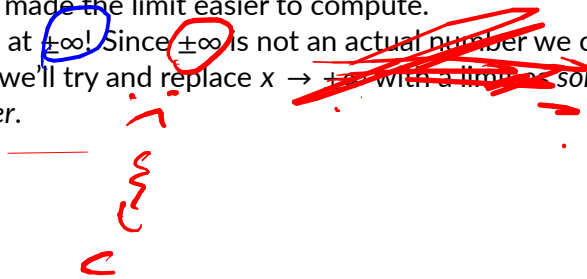


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Which allowed us to do a "substitution" that made the limit easier to compute.
This will be our strategy for computing limits at $\pm\infty$! Since $\pm\infty$ is not an actual number we can compute with in a consistent/coherent way, we'll try and replace $x \to \pm\infty$ with a limit of *some other quantity* approaches *some actual number*.



Limits at Infinity I

Before that, let's actually look at the literal definition to see why we want to avoid it:

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Do we want to prove things in this class like we would need to if we actually used this definition?

Limits at Infinity I



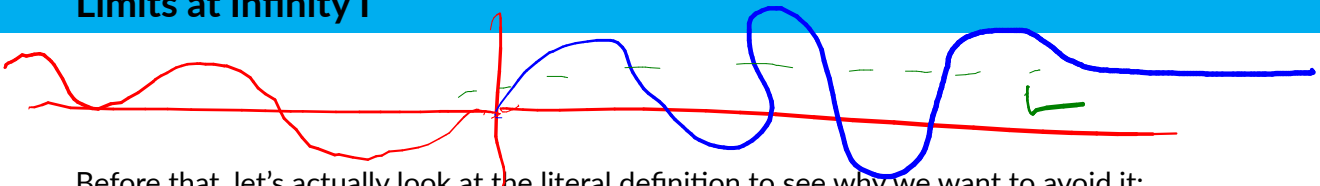
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For a function $f(x)$, we say that $\lim_{x\to\infty} f(x) = L$ if we can make $f(x) \approx L$ (up to some error $\epsilon > 0$) just by forcing $x > N$ for some ~~real number N.~~

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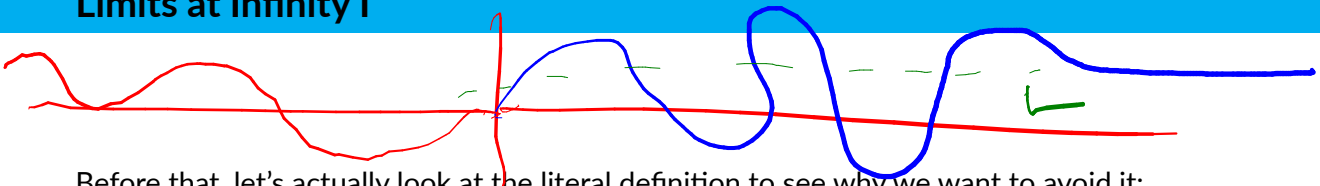
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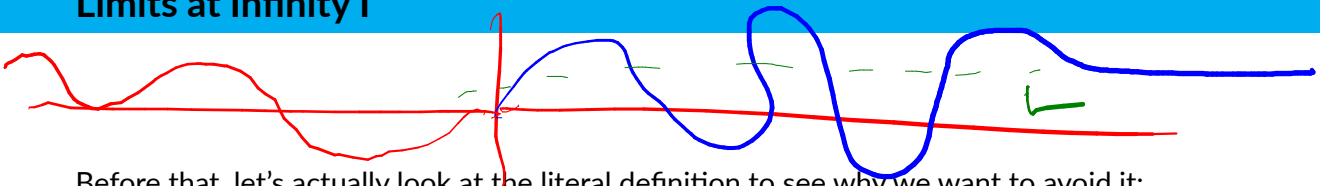
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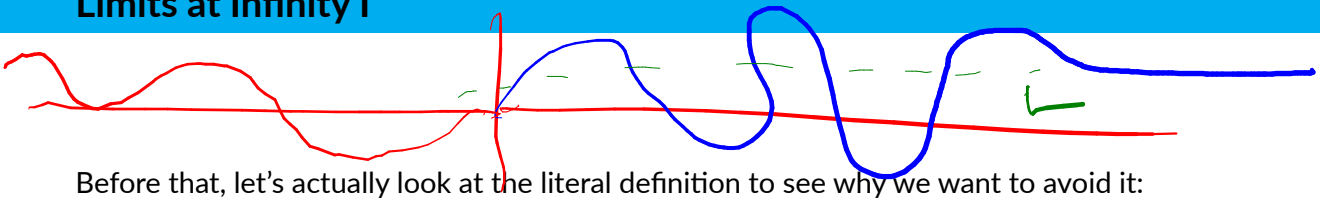
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Limits @ ∞ defn is equal for cont.

Limits at Infinity II

Instead, let's see if we can relate it to something we could instead *compute* using our tools from last lecture.

Limits at Infinity in terms of Finite Limits

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$\lim_{x \to \infty} g(x)$

$x > N \iff 0 > x > N$

Q: g

$x \to 0 \to 1/x \to \infty$

$? \approx c \in \mathbb{R}$

Limits at Infinity in terms of Finite Limits

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Instead, let's see if we can relate it to something we could instead *compute* using our tools from last lecture.

$$x > N \iff 0 > x > N \iff 0 < \frac{1}{x} < \frac{1}{N} \tag{2}$$

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In other words: $C \in \mathbb{R}$

Limits at Infinity in terms of Finite Limits

Taking the limit as $x \to +\infty$ is *exactly equivalent* to taking the limit as $\frac{1}{x} \to 0^+$

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Limits at Infinity in terms of Finite Limits

Taking the limit as $x \to +\infty$ is *exactly equivalent* to taking the limit as $\frac{1}{x} \to 0^+$

Similarly $x \to -\infty$ is *exactly equivalent* to $\frac{1}{x} \to 0^-$!

Limits at Infinity III: Example

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$u = 1/x$

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Limits at Infinity IV: Example 1+2

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What about for just a polynomial?

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$$\lim_{x\to-\infty} (x^3+2x+1) =$$

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$$\text{“}=\text{”}(\lim_{u\to 0^-}\frac{1}{u^3})\cdot(\lim_{u\to 0^-}(1+2u^2+u^3)) = 1$$

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Where we unfortunately encounter a problem since $\frac{1}{u^3}$ has an *asymptote* at $u = 0$

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Limits at Infinity V: Example 2 and Divergence to Infinity

Let's look at the graph!

Limits Diverging to Infinity

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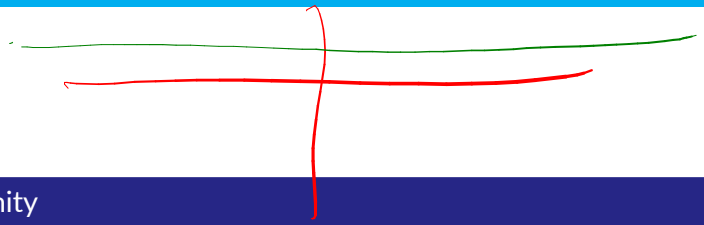
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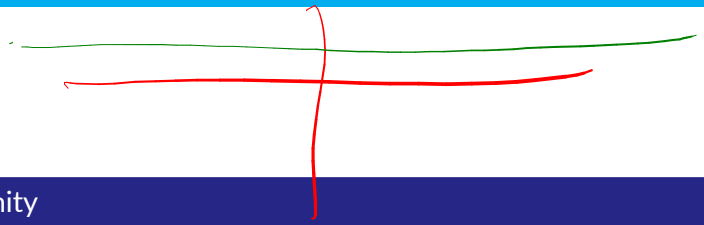
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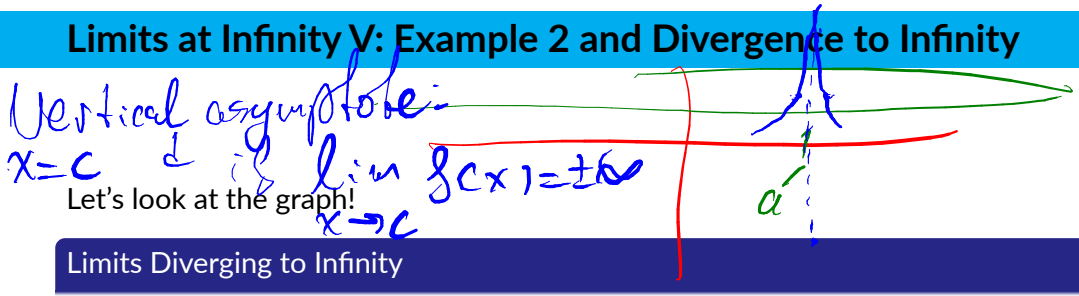
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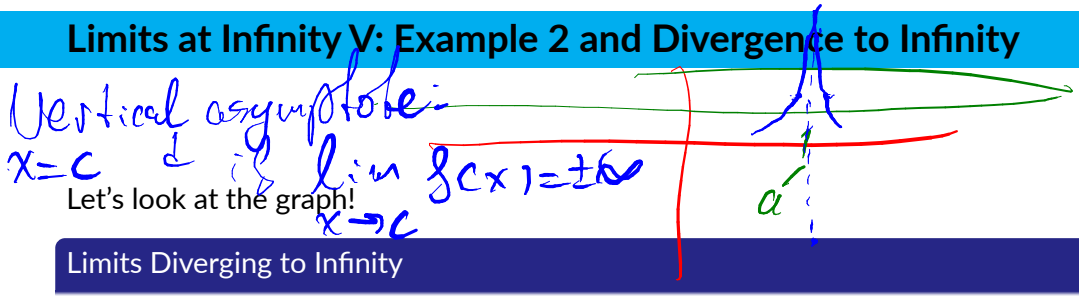
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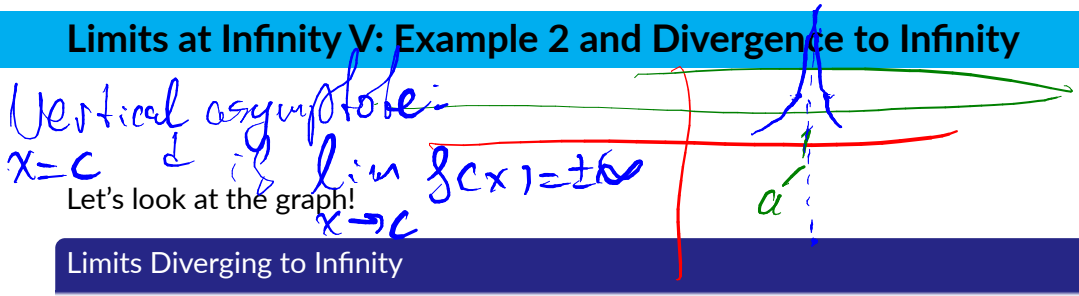
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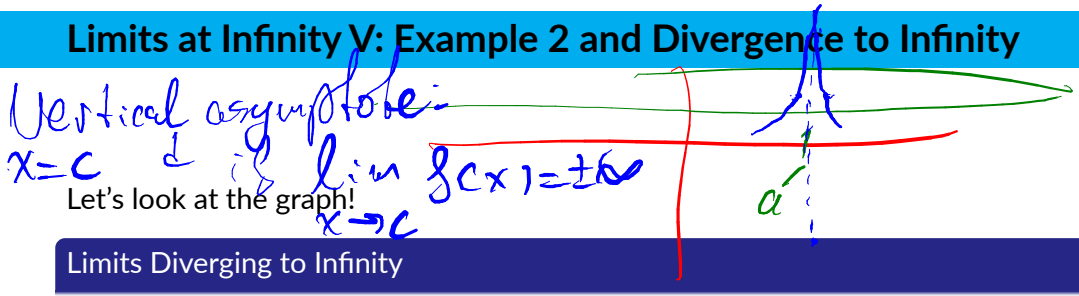
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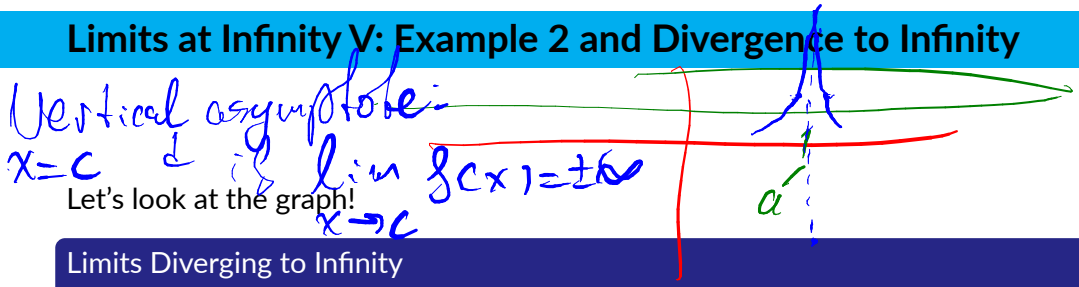
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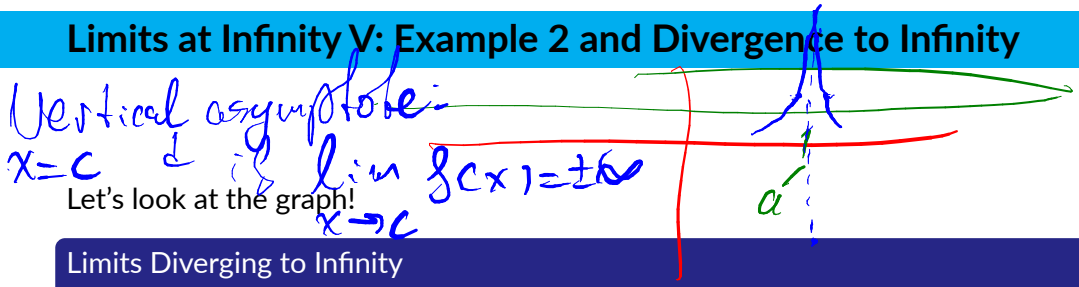
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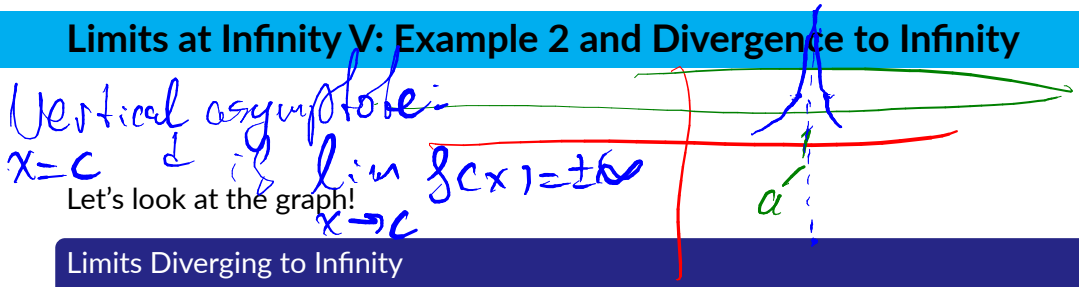
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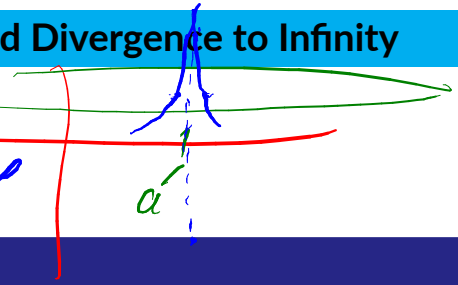
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Vertical asymptote: $x = c$ if $\lim_{x \to c} f(x) = \pm\infty$



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$\lim_{u \to -\infty} u^3 + 3u + 1 = -\infty$

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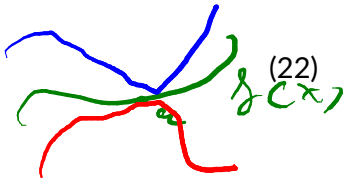
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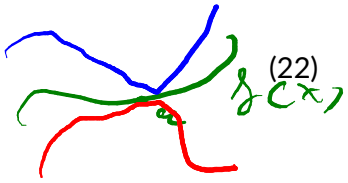
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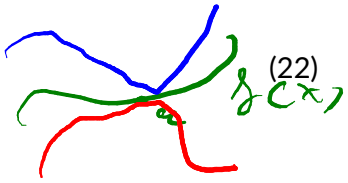
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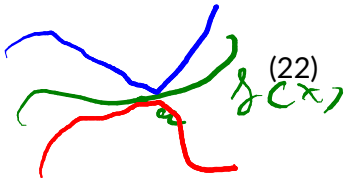
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Then we must have that $\lim_{x \to a} f(x)$ exists and is equal to the common value of $\lim_{x \to a} g(x) = \lim_{x \to a} h(x)$.

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Worksheet Time

Let’s get into groups and work on the worksheet!

End of Class

That’s all folks!